Spectral Properties of a Matrix of Redheffer

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ABSTRACT

Define $n \times n$ matrices $D_n = (d_{ij})$ and $C_n = (c_{ij})$ by $d_{ij} = 1$ if $i \mid j$, 0 otherwise, and $C_n = (0, 1, 1, \ldots, 1)^T(1, 0, 0, \ldots, 0)$. Let $A_n = D_n + C_n$. The matrix is of number-theoretic significance because $\det A_n = M(n)$, where $M(n)$ is Mertens' function. We give a simplified derivation of the characteristic polynomial of A_n, give precise asymptotic estimates for its two "large" eigenvalues, and use Rouché's theorem to significantly improve a previous bound on its "small" eigenvalues. Finally, we present some attractive conjectures about these small eigenvalues based on numerical evidence.

INTRODUCTION

The purpose of this paper is to give a simplified derivation, using a theorem from combinatorial matrix theory, of the characteristic polynomial of a matrix of Redheffer, to give precise asymptotic estimates for its two "large" eigenvalues, and to present a short argument giving a greatly improved bound on the moduli of its "small" eigenvalues. Our intention is to give an accessible account of what is currently known about the matrix and to state some challenging unresolved questions.

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First consider the semiinfinite matrix

$$D = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & \cdots \\ 0 & 1 & 0 & 1 & 0 & 1 & \cdots \\ 0 & 0 & 1 & 0 & 0 & 1 & \cdots \\ 0 & 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

and let A be the matrix created from D by changing the zeros in the first column to ones:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & \cdots \\ 1 & 1 & 0 & 1 & 0 & 1 & \cdots \\ 1 & 0 & 1 & 0 & 0 & 1 & \cdots \\ 1 & 0 & 0 & 1 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

Now define A_n to be the leading $n \times n$ principal submatrix of A. The matrix A_n is of number-theoretic interest because the Riemann hypothesis is true if and only if $|\det A_n| = O(n^{\frac{1}{2}+\varepsilon})$ for every positive ε, and a proof that $|\det A_n| = o(n)$ would constitute a new proof of the prime-number theorem. The connection to the Riemann hypothesis and the prime-number theorem is a result of the fact, proved by Redheffer when he first introduced A_n in [9], that

$$\det A_n = \sum_{k=1}^{n} \mu(k),$$

where $\mu : \mathbb{Z}^+ \to \{-1, 0, 1\}$ is the Möbius function. For more information about this sum, which is called Mertens' function, and its number-theoretic properties see Ivić [5].

THE CHARACTERISTIC POLYNOMIAL OF A_n

One way to derive the characteristic polynomial of A_n is to use the directed graph $D(M_n)$ of the matrix $M_n = \lambda I - A_n$, where $D(M_n)$ has vertex

set $N = \{1, 2, \ldots, n\}$ and a directed edge from vertex i to vertex j if and only if $m_{ij} \neq 0$. For convenience we will suppress the subscript n and write M wherever n is clear from context. In our case the graph $D(M)$ will have a directed edge from i to j if and only if $i \mid j$ or $j = 1$. We will denote a directed edge from i to j by the ordered pair (i, j). A set of directed edges $\{(i_1, j_1), (i_2, j_2), \ldots, (i_k, j_k)\}$ will be called a cycle of length k if $j_r = i_{r+1}$ for $r = 1, 2, \ldots, k-1$, $j_k = i_1$, and $i_1, i_2, \ldots, i_k$ are distinct. Furthermore, if $c = \{(i_1, i_2), (i_2, i_3), \ldots, (i_k, i_1)\}$ is a cycle in $D(M)$, then we set $P[c] = m_{i_1 i_2} m_{i_2 i_3} \cdots m_{i_k i_1}$, and for $\alpha \subseteq N$, let $M[\alpha]$ be the principal submatrix of M with rows and columns in α.

Let $\mathscr{C}_1$ be the set of cycles in $D(M)$ containing the vertex 1, and for each c in $\mathscr{C}_1$ let $l(c)$ be its length. Let c' be the complement (in N) of the vertices connected by c. Then by Theorem 2 of [8]

$$\det M = \sum_{c \in \mathscr{C}_1} (-1)^{l(c)+1} P[c] \det M[c']. \tag{1}$$

Notice that since each c in $\mathscr{C}_1$ contains the vertex 1, the principal submatrix $M[c']$ is upper triangular with determinant equal to $(\lambda - 1)^{n-l(c)}$. Furthermore, if c has length 2 or more, then $P[c]$ does not contain any terms from the diagonal and is exactly $(-1)^{l(c)}$. The only cycle of length 1 is $c = \{(1, 1)\}$, in which case $P[c] = \lambda - 1$. Thus

$$\det M_n = (\lambda - 1)^n - \sum_{\substack{c \in \mathscr{C}_1 \\ l(c) \geq 2}} (\lambda - 1)^{n-l(c)}, \tag{2}$$

which we can write as

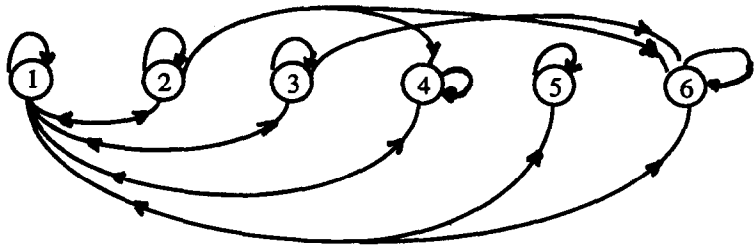
$$\det M_n = (\lambda - 1)^n - \sum_{k=2}^{n} (\lambda - 1)^{n-k} v_{n\,k-1}, \tag{3}$$

where v_{nk} is the number of cycles beginning at vertex 1, of length $k + 1$, in the graph $D(M_n)$. Equivalently, if a k-list in N is a list $\{1, i_1, i_2, \ldots, i_k\}$ with $i_1 \mid i_2 \mid \cdots \mid i_k$, and $1 < i_1 < i_2 < \cdots < i_k \leqslant n$, then v_{nk} is the number of distinct k-lists in N.

Applying this method to A_6, for example, we get

$$M_6 = \begin{bmatrix} \lambda-1 & -1 & -1 & -1 & -1 & -1 \\ -1 & \lambda-1 & 0 & -1 & 0 & -1 \\ -1 & 0 & \lambda-1 & 0 & 0 & -1 \\ -1 & 0 & 0 & \lambda-1 & 0 & 0 \\ -1 & 0 & 0 & 0 & \lambda-1 & 0 \\ -1 & 0 & 0 & 0 & 0 & \lambda-1 \end{bmatrix},$$

and $D(M_6)$ is



The cycles in $\mathscr{C}_1$ are

$\{(1,1)\}$;
$\{(1,2),(2,1)\}, \{(1,3),(3,1)\}, \{(1,4),(4,1)\}, \{(1,5),(5,1)\}, \{(1,6),(6,1)\}$;
$\{(1,2)(2,4),(4,1)\}, \{(1,2),(2,6),(6,1)\}, \{(1,3),(3,6),(6,1)\}$.

So there are five cycles of length two, three cycles of length three, and none of length greater than three. By Equation (3)

$$\det M_6 = (\lambda-1)^6 - 5(\lambda-1)^4 - 3(\lambda-1)^3.$$

In general, no cycle in the graph $D(M_n)$ can be longer than $[\log_2 n] + 1$, where $[x]$ is the greatest integer less than or equal to x. That is to say, there are no k-lists in N for $k > [\log_2 n]$. This can be seen most readily from the fact that the k-list in N having the smallest value for its last term must be $\{1, 2, 4, 8, \ldots, 2^k\}$; hence, if $n < 2^k$, there are no k-lists in N.

Let $s = [\log_2 n]$. We now have for the characteristic polynomial of A_n

$$P_{A_n}(\lambda) = \det M_n = (\lambda-1)^n - \sum_{k=1}^{s} v_{nk}(\lambda-1)^{n-k-1}. \tag{4}$$

From (4) it follows immediately that $\lambda = 1$ is an eigenvalue of algebraic

multiplicity $n - s - 1$. Our concern now will be with the factored polynomial

$$f_n(t) = \frac{P_{A_n}(t+1)}{t^{n-s-1}} = t^{s+1} - \sum_{k=1}^{s} v_{nk} t^{s-k}. \tag{5}$$

It will prove useful to compare $f_n(t)$ with the truncated polynomial $r_n(t) = t^{s+1} - v_{n1}t^{s-1}$. Since in N there is exactly one 1-list, $\{1, k\}$ for each $k \neq 1$, we have

$$v_{n1} = n - 1. \tag{6}$$

Thus

$$r_n(t) = t^{s+1} - (n-1)t^{s-1}, \tag{7}$$

which has roots $\sqrt{n-1}$, $-\sqrt{n-1}$, and 0 with multiplicity $s - 1$. We shall see that $f_n(t)$ also has a positive and a negative root, each of magnitude $\sqrt{n}$, and that the remaining $s - 1$ roots of $f_n(t)$ have much smaller modulus.

LARGE ROOTS OF f_n

The following result has been obtained by R. C. Vaughan.

THEOREM 1. *The polynomial $f_n(t)$ has two roots t_n^+, t_n^- which satisfy*

$$t_n^{\pm} = \pm\sqrt{n} + \log\sqrt{n} + \gamma - \frac{3}{2} + O\left(\frac{\log^2 n}{\sqrt{n}}\right) \tag{8}$$

as $n \to \infty$. Here $\gamma \approx 0.5772$ is Euler's constant.

Before giving a proof of Theorem 1, we compare it with previously known results. As noted in [2], $f_n(t)$ has exactly one positive root by Descartes' rule of signs. Hence this root is t_n^+. Since $1 + t_n^+ \in \sigma(A_n)$ and since $\rho(A_n)$ is an eigenvalue of A_n by Perron's theorem [4], we must have $\rho(A_n) = 1 + t_n^+$. Thus,

$$\rho(A_n) = \sqrt{n} + \log\sqrt{n} + \gamma - \frac{1}{2} + O\left(\frac{\log^2 n}{\sqrt{n}}\right), \tag{9}$$

which is a substantial improvement over the result $\rho(A_n) = \sqrt{n} + \log\sqrt{n} + O(1)$ announced at the end of [2]. An earlier bound for t_n^- given in Theorem 2 of [7] (there y_n is t_n^-) is also significantly improved by Theorem 1.

We need the following bound, proved in [2], for the positive integers v_{nk}:

$$v_{nk} < \frac{n \log^{k-1} n}{(k-1)!}. \tag{10}$$

Except for (10), and an asymptotic formula for v_{n2} given below, the following proof is self-contained.

Proof of Theorem 1. We have by (5), (6), and (10),

$$\left| \frac{f_n(t)}{t^{s-1}} - t^2 + n - 1 \right| \leqslant \sum_{k=2}^{s} v_{nk} |t|^{1-k} < \sum_{k=2}^{s} \frac{n}{(k-1)!} \left(\frac{\log n}{|t|} \right)^{k-1}$$

Therefore,

$$\left| \frac{f_n(t)}{t^{s-1}} - t^2 + n - 1 \right| < n \left\{ \exp\left(\frac{\log n}{|t|} \right) - 1 \right\}, \tag{11}$$

and similarly,

$$\left| \frac{f_n(t)}{t^{s-1}} - t^2 + n - 1 + \frac{v_{n2}}{t} \right| < n \left\{ \exp\left(\frac{\log n}{|t|} \right) - 1 - \frac{\log n}{|t|} \right\}. \tag{12}$$

We first establish the existence of $t_n^{\pm}$. Setting $t = c\sqrt{n}$ in (11),

$$\left| \frac{f_n(c\sqrt{n})}{(c\sqrt{n})^{s-1}} - c^2 n + n - 1 \right| < n \left\{ \exp\left(\frac{\log n}{|c|\sqrt{n}} \right) - 1 \right\},$$

so for n sufficiently large,

$$\left| \frac{f_n(c\sqrt{n})}{(c\sqrt{n})^{s-1}} - (c^2 - 1)n - 1 \right| < n \left(2 \frac{\log n}{|c|\sqrt{n}} \right) = \frac{2\sqrt{n} \log n}{|c|}. \tag{13}$$

Letting $c = \pm 2$ and $c = \pm \frac{1}{2}$, we see that for large n

$$\frac{f_n(\pm 2\sqrt{n})}{(\pm 2\sqrt{n})^{s-1}} > 0, \qquad \text{while} \quad \frac{f_n(\pm \frac{1}{2}\sqrt{n})}{(\pm \frac{1}{2}\sqrt{n})^{s-1}} < 0.$$

Thus, for n sufficiently large, $f_n(\pm 2\sqrt{n})f_n(\pm \frac{1}{2}\sqrt{n}) < 0$, so $f_n(t)$ has a root $t_n^+ \in (\frac{1}{2}\sqrt{n}, 2\sqrt{n})$ and another root $t_n^- \in (-2\sqrt{n}, -\frac{1}{2}\sqrt{n})$. Setting $t = t_n^\pm$ in (11), we have

$$\left|\left(t_n^\pm\right)^2 - (n-1)\right| < n\left\{\exp\left(\frac{\log n}{|t_n^\pm|}\right) - 1\right\},$$

so for large n,

$$\left|\left(t_n^\pm\right)^2 - n\right| < 2n\frac{\log n}{|t_n^\pm|} < 4\sqrt{n}\log n.$$

Then

$$\left|t_n^+ - \sqrt{n}\right| < \frac{4\sqrt{n}\log n}{t_n^+ + \sqrt{n}} = O(\log n)$$

and similarly, $|t_n^- + \sqrt{n}| = O(\log n)$. Thus, we have

$$t_n^\pm = \pm\sqrt{n} + O(\log n). \tag{14}$$

We now refine these estimates for $t_n^\pm$. Setting $t = t_n^\pm$ in (12), we have

$$\left|\left(t_n^\pm\right)^2 - (n-1) - \frac{v_{n2}}{t_n^\pm}\right| < n\left(\exp\left(\frac{\log n}{|t_n^\pm|}\right) - 1 - \frac{\log n}{|t_n^\pm|}\right).$$

Since the right-hand side is smaller than $(n\log^2 n)/|t_n^\pm|^2$ for large n, we have

$$\left(t_n^\pm\right)^2 = n + \frac{v_{n2}}{t_n^\pm} + O(\log^2 n). \tag{15}$$

It follows from (14) that

$$\frac{1}{t_n^\pm} = \pm\frac{1}{\sqrt{n}} + O\left(\frac{\log n}{n}\right),$$

so by (10)

$$\frac{v_{n2}}{t_n^{\pm}} = \pm \frac{v_{n2}}{\sqrt{n}} + O\left(\log^2 n\right).$$

Thus by (15),

$$\left(t_n^{\pm}\right)^2 = n \pm \frac{v_{n2}}{\sqrt{n}} + O\left(\log^2 n\right)$$

$$= n\left\{1 \pm \frac{v_{n2}}{n^{3/2}} + O\left(\frac{\log^2 n}{n}\right)\right\}.$$

Since $\sqrt{1+x} = 1 + \frac{1}{2}x + O(x^2)$ as $x \to 0$, and since by (10) $v_{n2}/n^{3/2} < (\log n)/\sqrt{n}$, we have

$$t_n^{\pm} = \pm\sqrt{n}\left\{1 \pm \frac{v_{n2}}{2n^{3/2}} + O\left(\frac{\log^2 n}{n}\right)\right\}$$

$$= \pm\sqrt{n} + \frac{v_{n2}}{2n} + O\left(\frac{\log^2 n}{\sqrt{n}}\right). \tag{16}$$

Now, by definition,

$$v_{n2} = \sum_{j=1}^{n} \sum_{\substack{i \mid j \\ 1<i<j}} 1 = \sum_{j=2}^{n} \left[d(j) - 2\right] = \sum_{j=1}^{n} d(j) - 2n + 1,$$

where $d(j)$ is the number of divisors of j. Then, by Theorem 3.3 in [1],

$$v_{n2} = n \log n + (2\gamma - 3)n + O\left(\sqrt{n}\right).$$

Substituting in (16) yields

$$t_n^{\pm} = \pm\sqrt{n} + \tfrac{1}{2}\log n + \gamma - \frac{3}{2} + O\left(\frac{\log^2 n}{\sqrt{n}}\right),$$

which is (8). ■

BOUND ON THE REMAINING ROOTS OF f_n

All of the remaining roots of f_n have relatively small moduli, as indicated in the following theorem. This theorem represents a substantial improvement over the bound given in Jarvis [7] for these small roots.

THEOREM 2. *Given $\varepsilon > 0$, for large enough n, f_n has $s - 1$ roots inside the circle*

$$\{z \in \mathbb{C} : |z| < \log_{2-\varepsilon} n\}.$$

Proof. We have the following bound for $r_n(t) = t^{s+1} - v_{n1}t^{s-1}$ on the circle $|t| = \log_{2-\varepsilon} n$:

$$\begin{aligned} |r_n(t)| &= (\log_{2-\varepsilon} n)^{s-1} |v_{n1} - t^2| \\ &\geqslant (\log_{2-\varepsilon} n)^{s-1} (n - 1 - \log_{2-\varepsilon}^2 n). \end{aligned}$$

From the bounds of (10) and the triangle inequality,

$$\begin{aligned} |f_n(t) - r_n(t)| &= (\log_{2-\varepsilon} n)^{s-1} \left| \sum_{k=2}^{s} \frac{v_{nk}}{t^{k-1}} \right| \\ &< (\log_{2-\varepsilon} n)^{s-1} \sum_{k=2}^{s} \frac{n \log^{k-1} n}{(k-1)!(\log_{2-\varepsilon} n)^{k-1}} \\ &< (\log_{2-\varepsilon} n)^{s-1} n(e^{\log(2-\varepsilon)} - 1) \\ &= (1 - \varepsilon) n (\log_{2-\varepsilon} n)^{s-1} \\ &< (n - 1 - \log_{2-\varepsilon}^2 n)(\log_{2-\varepsilon} n)^{s-1} \leqslant |r_n(t)| \end{aligned}$$

for sufficiently large n.

So by Rouché's theorem $r_n(t)$ and $f_n(t)$ have the same number of zeros in the circle $\{z \in \mathbb{C} : |z| < \log_{2-\varepsilon} n\}$. ∎

We end with the following two conjectures:

CONJECTURE 1. All but two of the roots of $f_n(t)$ lie in the circle $\{z \in \mathbb{C} : |z + 1| \leqslant 1\}$.

CONJECTURE 2. All roots of $f_n(t)$ except t_n^+ have negative real part.

Of course Conjecture 1 implies Conjecture 2. Both have been verified in numerical trials for all values of $n \leqslant 500$ and for selected values as large as one million. Figure 1, which was generated in Mathematica, shows a plot of 18

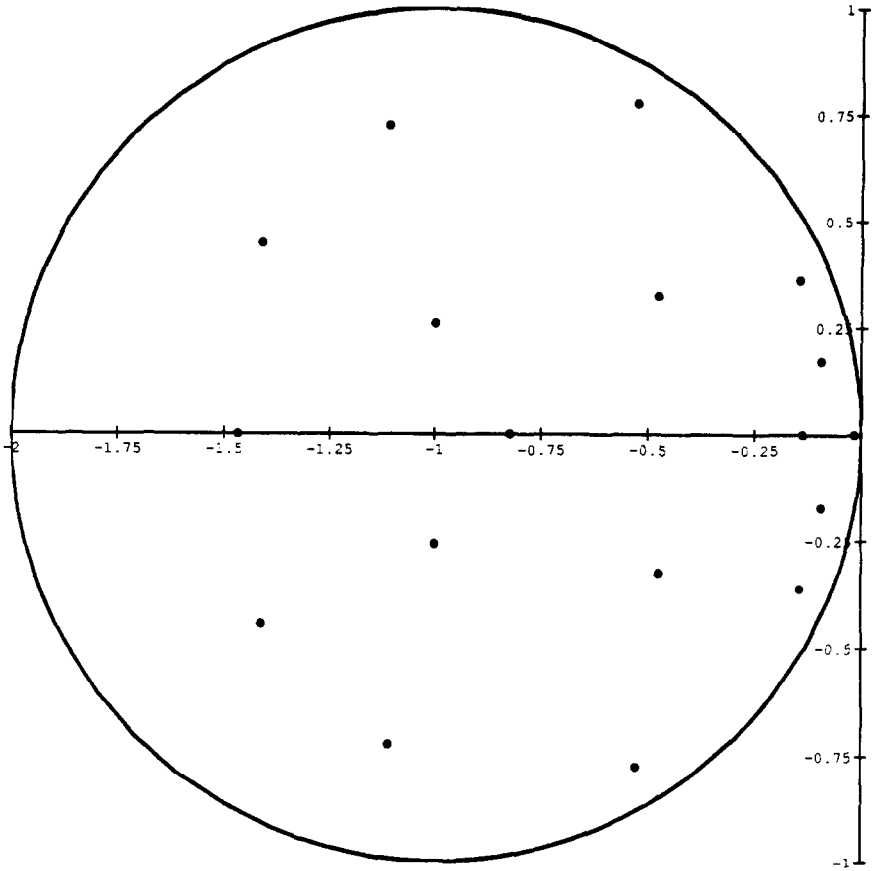


FIG. 1.

of the 20 roots of $f_n(t)$ for $n = 10^6$. We find this plot remarkable, considering the fact that the two large roots are approximately 1006 and -994.

Since the sum of all roots of $f_n(t)$ is 0, by Theorem 1 the sum of the $s - 1$ roots of "small" modulus is $3 - 2\gamma - \log n + o(1)$. Dividing by $s - 1 = [\log_2 n] - 1$, we see that the average of the $s - 1$ "small" roots approaches $-\log 2 \approx -0.693$ as $n \to \infty$. So at least the average of the "small" roots lies in the circle $|z + 1| \leqslant 1$ for large n. We still feel the first conjecture may fail for very large n, but are more confident in the validity of the second.

In conclusion, we know that A_n has $n - s - 1$ eigenvalues equal to 1, has spectral radius $\rho(A_n) \sim \sqrt{n}$, has a negative eigenvalue $t_n^- + 1 \sim -\sqrt{n}$, and has $s - 1$ other eigenvalues $\lambda_1, \ldots, \lambda_{s-1}$, each of which is at most $O(\log_{2-\varepsilon} n)$. In particular, $|\det A_n| = O(n \prod_{k=1}^{s-1} \lambda_k)$. Note that Conjecture 1 is equivalent

to $\lambda_1, \ldots, \lambda_{s-1}$ all lying inside the circle $|z| = 1$; consequently, it has considerable number-theoretic significance. For example, a proof of the prime-number theorem could be given by establishing Conjecture 1 and, in addition, showing that some eigenvalue of A_n tends to 0 as $n \to \infty$.

We thank R. C. Vaughan of Imperial College, London, for his valuable assistance with this paper. He not only improved upon our asymptotic estimates for $t_n^{\pm}$ by giving the next term, $\gamma - \frac{3}{2}$ in each, but achieved a great simplification over our results by unifying the argument for t_n^+ and t_n^-. The above proof of Theorem 1 is an amplified version of one he sent us. Finally, he suggested that a logarithmic bound on the small roots of f_n should be attainable from Rouché's theorem.

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